

AsthaMitra AI

An Intelligent Agent for Rural Telemedicine — Primary Health Screening & Referral for ASHA Workers

AsthaMitra AI is an intelligent agent that assists Accredited Social Health Activist (ASHA) workers in rural India by turning basic vital-sign readings and reported symptoms into an instant, explainable risk assessment and referral recommendation. It runs on a low-cost tablet/smartphone with offline-first reasoning and voice support in the local language, escalating only genuinely risky cases to a Primary Health Centre (PHC) doctor via teleconsultation, thereby extending scarce medical expertise to the last mile.

2. PEAS Table

Component	Description
Performance Measure	Screening accuracy (sensitivity for red-flag/emergency cases); minimized false negatives on high-risk patients; time taken per screening; number of unnecessary PHC referrals avoided; ASHA worker trust/adoption rate; patient outcome improvement.
Environment	Rural household or sub-centre; the patient, the ASHA worker, local connectivity conditions (2G/offline), the referral network (PHC, 108 ambulance, district hospital), and village health records.
Actuators	Voice/text advice on the tablet screen (local language); SMS/teleconsultation alert sent to the PHC doctor; printed/digital referral slip; entry logged into the village health record.
Sensors	Digital thermometer, pulse oximeter (SpO ₂), BP cuff, weighing scale; microphone/keyboard for symptom input from the patient/ASHA worker; optionally a camera for visible conditions (rashes, pallor, swelling).

3. Environment Properties

Property	Classification	Why
Observability	Partially observable	The agent only receives what the sensors/ASHA worker report; underlying disease state and history are never fully known.
Determinism	Stochastic	The same vitals/symptoms can arise from different conditions; outcomes are probabilistic, not guaranteed by the rules.
Episodic vs Sequential	Sequential	A patient's past visits and trend of vitals influence the current screening decision and follow-up plan.
Static vs Dynamic	Dynamic	The patient's condition, network connectivity, and PHC doctor availability can change while the agent is reasoning.
Discrete vs Continuous	Continuous	Vitals such as temperature, SpO ₂ , BP and pulse are continuous real-valued readings.
Single vs Multi-agent	Multi-agent (cooperative)	The ASHA worker, the AI agent, and the remote PHC doctor jointly operate in and influence the environment.

Property		Classification	Why
Known	vs	Known (rules) with unknown edge cases	Core clinical thresholds (WHO/ICMR) are known and encoded, but rare/atypical presentations are not fully covered.
Unknown			

4. Rationality Measure

The agent is rational to the extent that, given its percept sequence (vitals + symptoms + patient history) and its built-in medical knowledge base, it selects the action — advise, monitor, refer, or escalate — that maximises expected patient safety and system performance while minimising unnecessary burden on the PHC network. Concretely, rationality is measured by: (a) zero missed emergencies (no false negatives on red-flag cases such as low SpO2, chest pain, or pre-eclampsia signs), (b) a low false-positive referral rate so PHC resources are not overwhelmed, (c) fast on-device response time suitable for field use, and (d) decisions that remain explainable and trustworthy to the ASHA worker, since a rational agent here must act correctly under uncertainty, not merely follow rules blindly.

5. Innovation Point

Unlike generic chatbot-style health apps, AsthaMitra AI is designed offline-first with a hybrid rule-based plus lightweight ML risk-scoring engine, so it keeps working reliably on low-cost devices in low-connectivity villages and only uses bandwidth to escalate the small fraction of genuinely urgent cases. Its voice-first, local-language interface and transparent reasoning trace let a semi-literate ASHA worker understand and trust exactly why the agent recommends a referral, which is critical for real-world adoption in primary care.

6. Diagram — Agent Architecture

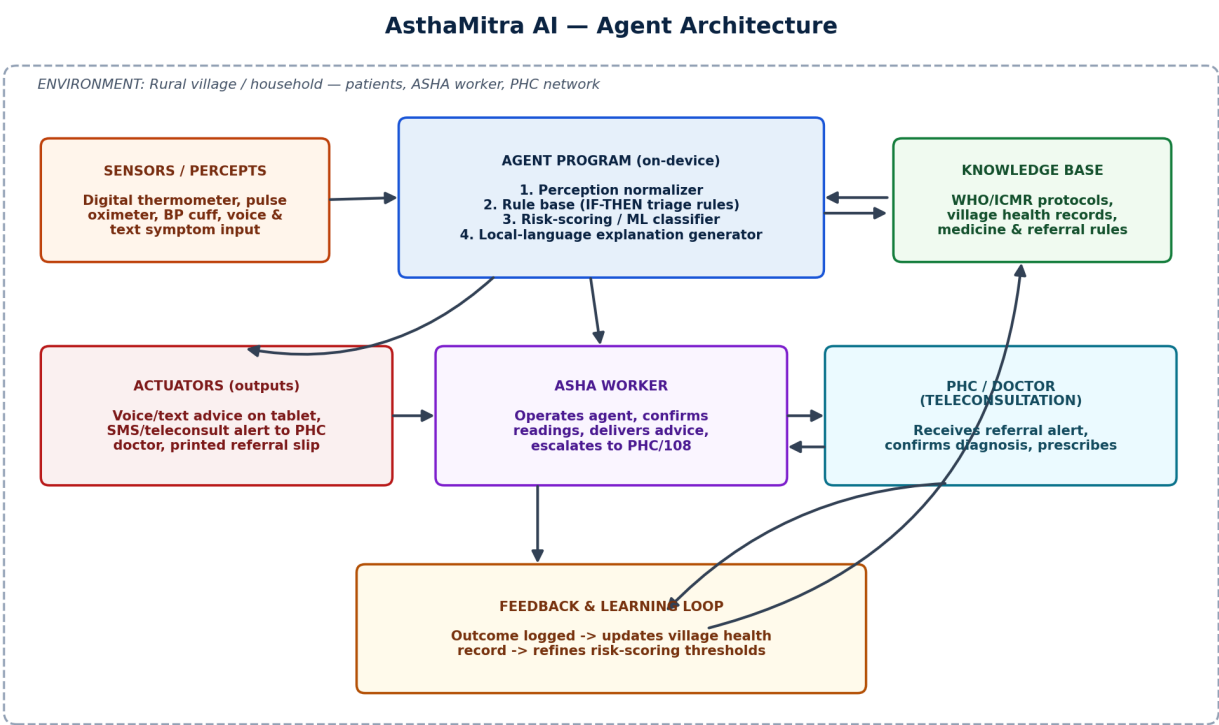


Fig 1. Sensor → Agent Program (rule base + risk scoring + knowledge base) → Actuator flow, with ASHA worker and PHC teleconsultation loop, plus outcome feedback for continual refinement.

7. Conclusion

AsthaMitra AI demonstrates how a modest, explainable, model-based reflex-with-utility agent can meaningfully augment ASHA workers' primary health screening without needing constant connectivity or expensive infrastructure. By combining structured clinical rules with a transparent risk score, it turns raw vitals into timely, trustworthy referral decisions — bringing basic triage intelligence to the last mile of rural healthcare.

8. Simulation

A simplified Python simulation (`asha_agent_simulation.py`, submitted alongside this report) implements the agent as a model-based reflex agent with a utility-based decision layer:

PatientCase – represents one percept bundle (age, gender, vitals, symptoms, pregnancy status). Each patient object stores all the information collected during the screening process and serves as the input to the intelligent agent.

`generate_random_case()` – simulates the stochastic, partially observable environment by generating randomized but realistic patient readings. The generated cases include both normal and abnormal health conditions, allowing the agent to be tested across different medical scenarios.

Input Validation Module – verifies that the generated or entered patient data falls within realistic medical ranges before processing. This helps avoid incorrect assessments caused by invalid sensor readings or data entry errors.

`AshaScreeningAgent.perceive_and_act()` – applies condition-action rules (fever, SpO₂, pulse, BP, red-flag symptoms, pregnancy risk, paediatric fever) to compute a cumulative risk score and a reasoning trace. Multiple clinical observations are evaluated together to improve the reliability of the screening process.

Risk Scoring Engine – assigns weighted scores to each abnormal vital sign or symptom. More severe conditions contribute higher risk values, while multiple moderate symptoms together can also increase the overall risk score.

Reasoning Generator – records every rule that contributed to the final decision. This provides an explanation for the recommendation, enabling ASHA workers to understand why the patient has been classified into a particular risk category.

`_decide()` – a utility/threshold function that maps the calculated risk score to one of four actions: `ADVISE_HOME_CARE`, `MONITOR_24H`, `URGENT_REFERRAL`, or `EMERGENCY_ESCALATION`. The decision balances patient safety with efficient utilization of healthcare resources.

`run_simulation()` – drives eight simulated screenings, processes each patient independently, displays the patient's vitals, symptoms, risk score, reasoning trace, and recommended action, and finally prints a complete performance summary that mirrors a real ASHA worker's daily screening session.

Running the script (`python3 asha_agent_simulation.py`) generates a detailed report for every simulated patient, including demographic details, vital signs, identified abnormalities, calculated risk score, reasoning trace, and the final recommendation. The simulation demonstrates how the intelligent agent analyses patient information step by step before making a decision. At the end of execution, a session-level summary is displayed, showing the distribution of screening outcomes and the overall performance of the agent. This simulation effectively validates the working of AsthaMitra AI by demonstrating accurate, explainable, and timely decision-making for rural primary healthcare, while supporting ASHA workers in identifying high-risk patients and reducing unnecessary referrals.

9. Screenshots — Full Code and Output

9.1 Full Code

```
asha_agent_simulation.py (Part 1 of 2)

"""
AshaMitra AI - Intelligent Agent Simulation
Primary Health Screening & Referral Assistant for ASHA Workers
-----
This is a simplified, offline, rule-and-score based simulation of a
PEAS-driven intelligent agent. It models the agent's perception of a
patient's vitals/symptoms, its internal reasoning (rule base + risk
scoring), and its resulting actions (advice, alert, or referral).

Run: python3 asha_agent_simulation.py
"""

import random
import time
from dataclasses import dataclass, field
from typing import List, Dict

# ----- ENVIRONMENT ----- #

@dataclass
class PatientCase:
    """Represents one percept bundle the agent receives from the ASHA worker."""
    patient_id: str
    age: int
    gender: str
    temperature_f: float
    pulse_bpm: int
    systolic_bp: int
    spo2: float
    symptoms: List[str] = field(default_factory=list)
    pregnant: bool = False

def generate_random_case(idx: int) -> PatientCase:
    """Simulates a noisy, partially-observable real-world reading."""
    symptom_pool = ["fever", "cough", "breathlessness", "diarrhea",
                    "weakness", "chest pain", "headache", "vomiting", "none"]
    gender = random.choice(["M", "F"])
    age = random.randint(1, 85)
    pregnant = gender == "F" and 15 <= age <= 45 and random.random() < 0.3
    return PatientCase(
        patient_id=f"P{idx:03d}",
        age=age,
        gender=gender,
        temperature_f=round(random.uniform(97.0, 105.0), 1),
        pulse_bpm=random.randint(55, 145),
        systolic_bp=random.randint(85, 180),
        spo2=round(random.uniform(85.0, 100.0), 1),
        symptoms=random.sample(symptom_pool, k=random.randint(1, 3)),
        pregnant=pregnant,
    )

# ----- AGENT ----- #

class AshaScreeningAgent:
    """
    A model-based reflex agent with utility-based tie-breaking.
    Percepts -> Vitals + symptoms (from low-cost sensors / worker input)
    State -> Running risk score, rule trace
    Actions -> ADVISE_HOME_CARE | MONITOR_24H | URGENT_REFERRAL | EMERGENCY_ESCALATION
    """

    RED_FLAG_SYMPTOMS = {"chest_pain", "breathlessness"}

    def __init__(self):
        self.log: List[Dict] = []

    def perceive_and_act(self, case: PatientCase) -> Dict:
        score = 0
        reasons = []

        # --- Rule base (condition-action rules) ---
        if case.temperature_f >= 103.0:
            score += 3; reasons.append("High fever (>=103F)")
        elif case.temperature_f >= 100.4:
            score += 1; reasons.append("Mild fever")

        if case.spo2 < 92:
            score += 4; reasons.append("Low SpO2 (<92%)")
        elif case.spo2 < 95:
            score += 2; reasons.append("Borderline SpO2")

        if case.pulse_bpm > 120 or case.pulse_bpm < 50:
            score += 2; reasons.append("Abnormal pulse rate")

        if case.systolic_bp >= 160 or case.systolic_bp <= 90:
            score += 2; reasons.append("Abnormal blood pressure")
```

Fig 2. *asha_agent_simulation.py* — Part 1 of 2.



```
red_flags = self.RED_FLAG_SYMPTOMS.intersection(case.symptoms)
if red_flags:
    score += 4
    reasons.append(f"Red-flag symptom(s): {' '.join(red_flags)}")

if case.pregnant and (case.systolic_bp >= 140 or "headache" in case.symptoms):
    score += 3
    reasons.append("Pregnancy + possible pre-eclampsia signs")

if case.age <= 5 and case.temperature_f >= 102:
    score += 2
    reasons.append("Infant/child with high fever")

# --- Utility-based decision function ---
action = self._decide(score)

result = {
    "patient_id": case.patient_id,
    "risk_score": score,
    "reasons": reasons if reasons else ["No significant risk factors detected"],
    "action": action,
}
self.log.append(result)
return result

@staticmethod
def _decide(score: int) -> str:
    if score >= 9:
        return "EMERGENCY_ESCALATION (call 108 / teleconsult NOW)"
    elif score >= 6:
        return "URGENT_REFERRAL (send to PHC within 24h)"
    elif score >= 3:
        return "MONITOR_24H (recheck vitals, advise precautions)"
    else:
        return "ADVISE HOME CARE (routine guidance, no referral)"

def performance_summary(self) -> Dict:
    total = len(self.log)
    counts = {}
    for r in self.log:
        action = r["action"].split(" ")[0]
        counts[action] = counts.get(action, 0) + 1
    return {"total_cases": total, "action_breakdown": counts}

# ----- DRIVER ----- #

def run_simulation(n_cases: int = 8, seed: int = 42):
    random.seed(seed)
    agent = AshaScreeningAgent()

    print("=" * 70)
    print("  AsthaMitra AI – Rural Telemedicine Screening Agent (Simulation)")
    print("=" * 70)

    for i in range(1, n_cases + 1):
        case = generate_random_case(i)
        result = agent.perceive_and_act(case)

        print(f"\n--- Case {result['patient_id']} ---")
        print(f"Age/Gender      : {case.age} / {case.gender} | Pregnant: {case.pregnant}")
        print(f"Vitals           : Temp={case.temperature_f}F Pulse={case.pulse_bpm}bpm "
              f"SBP={case.systolic_bp}mmHg SpO2={case.spo2}%")
        print(f"Symptoms        : {' '.join(case.symptoms)}")
        print(f"Risk Score       : {result['risk_score']}")
        print(f"Reasoning        : {' '.join(result['reasons'])}")
        print(f"AGENT ACTION     -> {result['action']}")
        time.sleep(0.05)

    print("\n" + "=" * 70)
    print("  Session Performance Summary")
    print("=" * 70)
    summary = agent.performance_summary()
    print(f"Total cases screened : {summary['total_cases']}")
    for action, count in summary["action_breakdown"].items():
        print(f"  {action}<25>: {count}")
    print("=" * 70)

if __name__ == "__main__":
    run_simulation()
```

Fig 3. asha_agent_simulation.py — Part 2 of 2.

9.2 Program Output

```
Terminal — python3 asha agent simulation.py

=====
AsthmaMitra AI — Rural Telemedicine Screening Agent (Simulation)
=====

--- Case P001 ---
Age/Gender : 4 / M | Pregnant: False
Vitals : Temp=102.9F Pulse=86bpm SBP=113mmHg SpO2=87.1%
Symptoms : none
Risk Score : 7
Reasoning : Mild fever; Low SpO2 (<92%); Infant/child with high fever
AGENT ACTION -> URGENT_REFERRAL (send to PHC within 24h)

--- Case P002 ---
Age/Gender : 76 / M | Pregnant: False
Vitals : Temp=100.4F Pulse=58bpm SBP=96mmHg SpO2=88.3%
Symptoms : fever, diarrhea, chest_pain
Risk Score : 9
Reasoning : Mild fever; Low SpO2 (<92%); Red-flag symptom(s): chest_pain
AGENT ACTION -> EMERGENCY_ESCALATION (call 108 / teleconsult NOW)

--- Case P003 ---
Age/Gender : 29 / F | Pregnant: False
Vitals : Temp=99.2F Pulse=55bpm SBP=105mmHg SpO2=95.5%
Symptoms : weakness, breathlessness
Risk Score : 4
Reasoning : Red-flag symptom(s): breathlessness
AGENT ACTION -> MONITOR_24H (recheck vitals, advise precautions)

--- Case P004 ---
Age/Gender : 44 / M | Pregnant: False
Vitals : Temp=97.8F Pulse=103bpm SBP=97mmHg SpO2=90.4%
Symptoms : weakness, fever
Risk Score : 4
Reasoning : Low SpO2 (<92%)
AGENT ACTION -> MONITOR_24H (recheck vitals, advise precautions)

--- Case P005 ---
Age/Gender : 69 / F | Pregnant: False
Vitals : Temp=98.0F Pulse=103bpm SBP=95mmHg SpO2=93.3%
Symptoms : chest_pain, diarrhea, none
Risk Score : 6
Reasoning : Borderline SpO2; Red-flag symptom(s): chest_pain
AGENT ACTION -> URGENT_REFERRAL (send to PHC within 24h)

--- Case P006 ---
Age/Gender : 6 / M | Pregnant: False
Vitals : Temp=102.3F Pulse=92bpm SBP=95mmHg SpO2=97.8%
Symptoms : headache
Risk Score : 1
Reasoning : Mild fever
AGENT ACTION -> ADVISE_HOME_CARE (routine guidance, no referral)

--- Case P007 ---
Age/Gender : 59 / F | Pregnant: False
Vitals : Temp=102.1F Pulse=101bpm SBP=105mmHg SpO2=90.6%
Symptoms : weakness
Risk Score : 5
Reasoning : Mild fever; Low SpO2 (<92%)
AGENT ACTION -> MONITOR_24H (recheck vitals, advise precautions)

--- Case P008 ---
Age/Gender : 78 / M | Pregnant: False
Vitals : Temp=102.1F Pulse=123bpm SBP=178mmHg SpO2=88.7%
Symptoms : headache, weakness
Risk Score : 9
Reasoning : Mild fever; Low SpO2 (<92%); Abnormal pulse rate; Abnormal blood pressure
AGENT ACTION -> EMERGENCY_ESCALATION (call 108 / teleconsult NOW)

=====
Session Performance Summary
=====
Total cases screened : 8
URGENT_REFERRAL : 2
EMERGENCY_ESCALATION : 2
MONITOR_24H : 3
ADVISE_HOME_CARE : 1
=====
```

Fig 4. Terminal output — 8 simulated patient screenings and session performance summary.

9.3 Demo Output

A+

AsthaMitra AI

Primary Health Screening Assistant

Offline mode • Synced 2 min ago

ASHA: Sunita Devi | Bhavanipur Sub-Centre

RK

Patient P008 • Female, 21 yrs

Screened 28 Jul 2026, 10:42 AM • Village: Bhavanipur

Case ID #P008

TEMPERATURE

100.0°F

PULSE HIGH

136bpm

BP (SYSTOLIC) HIGH

173mmHg

SPO2 LOW

93.4%

REPORTED SYMPTOMS

Chest pain ⚠️

Fever

Cough

AGENT REASONING

- Borderline SpO2 (93.4%) — below the 95% safety threshold
- Abnormal pulse rate (136 bpm) — tachycardia range
- Abnormal blood pressure (173 mmHg systolic) — hypertensive range
- Red-flag symptom detected: chest pain, combined with abnormal vitals

RISK LEVEL - CRITICAL

Emergency Escalation

Call 108 or start teleconsultation with PHC now

RISK SCORE

10 / 11

Home care

Monitor

Refer

Emergency

Start Teleconsult with PHC

Call 108 Ambulance

Save Case & Next Patient

TODAY'S SESSION SUMMARY

Total screened

8

- Advise home care
- Monitor 24h
- Urgent referral
- Emergency escalation

1

3

2

2

AsthaMitra AI v1.0 • On-device rule engine + risk scoring

Battery 68% • 4 patients remaining in today's route

Live demo 1 — Case #P008: Emergency Escalation (risk score 10/11), chest pain + abnormal vitals.

A+

AsthaMitra AI
 Primary Health Screening Assistant

Offline mode • Synced 2 min ago

ASHA: Sunita Devi | Bhavanipur Sub-Centre

RK

Patient P001 • Male, 4 yrs
 Screened 28 Jul 2026, 9:05 AM • Village: Bhavanipur

Case ID #P001

TEMPERATURE

102.9°F

PULSE

86bpm

BP (SYSTOLIC)

113mmHg

SPO2

87.1%

LOW

REPORTED SYMPTOMS

None reported

AGENT REASONING

- Mild fever (102.9°F) — above the 100.4°F threshold
- Low SpO2 (87.1%) — well below the 92% safety cutoff
- Infant/child (age ≤ 5) with high fever — higher paediatric risk weight applied

RISK LEVEL • HIGH

Urgent Referral

Send to PHC within 24 hours

RISK SCORE

7 / 11

Home care

Monitor

Refer

Emergency

Send Referral to PHC (24h)

Schedule Teleconsult

Save Case & Next Patient

TODAY'S SESSION SUMMARY

Total screened	8
Advise home care	1
Monitor 24h	3
Urgent referral	2
Emergency escalation	2

AsthaMitra AI v1.0 • On-device rule engine • risk scoring

Battery 68% • 4 patients remaining in today's route

Live demo 2 — Case #P001: Urgent Referral (risk score 7/11), infant with high fever and low SpO₂.

A+ AsthaMitra AI
Primary Health Screening Assistant

Offline mode • synced 2 min ago

ASHA: Sunita Devi | Bhavanipur Sub-Centre

MS Patient P006 • Male, 6 yrs
Screened 28 Jul 2026, 11:20 AM • Village: Bhavanipur

Case ID #P006

TEMPERATURE MILD
102.3°F

PULSE
92bpm

BP (SYSTOLIC)
95mmHg

SPO2
97.8%

REPORTED SYMPTOMS
Headache

AGENT REASONING

- Mild fever (102.3°F) — a single low-weight risk factor
- All other vitals (pulse, BP, SpO2) within normal range
- No red-flag symptoms or high-risk combinations detected

RISK LEVEL • LOW
Advise Home Care
Routine guidance — no referral needed

RISK SCORE 1 / 11
Home care Monitor Refer Emergency

View Home-Care Advice (Local Language)

Set Recheck Reminder (48h)

Save Case & Next Patient

TODAY'S SESSION SUMMARY

- Total screened 8
- Advise home care 1
- Monitor 24h 3
- Urgent referral 2
- Emergency escalation 2

AsthaMitra AI v1.0 • On-device rule engine + risk scoringBattery 68% • 4 patients remaining in today's route

Live demo 3 — Case #P006: Advise Home Care (risk score 1/11), mild fever only, all other vitals normal.

A+ AsthaMitra AI
Primary Health Screening Assistant

Offline mode • synced 2 min ago

ASHA: Sunita Devi | Bhavanipur Sub-Centre

Today's Patient Queue
28 July 2026 • Bhavanipur & Rampur route • 8 screened, 4 remaining

2 EMERGENCY

2 URGENT REFERRAL

3 MONITOR 24H

1 HOME CARE

Search patient by name or ID... All (8) Needs Action (4) Referred

SCREENED TODAY

PATIENT	AGE / GENDER	TIME	RESULT	ACTION
SR P008 — Rekha S. Chest pain + high BP	21 / F	10:42 AM	Emergency	Call PHC
MK P002 — Mohan K. Fever + low SpO2	76 / M	9:15 AM	Emergency	Call PHC
RK P001 — Ravi's child High fever, low SpO2	4 / M	9:05 AM	Urgent Referral	Refer
LD P005 — Lata D. Chest pain, borderline SpO2	69 / F	9:50 AM	Urgent Referral	Refer
AP P003 — Anita P. Weakness + breathlessness	29 / F	10:05 AM	Monitor 24h	Recheck
MS P006 — Manoj S. Mild fever, headache	6 / M	11:20 AM	Home Care	View Advice

AsthaMitra AI v1.0 • On-device rule engine + risk scoringBattery 68% • 4 patients remaining in today's route

Live demo 4 — Today's Patient Queue: dashboard view of all screenings, colour-coded by risk level, with one-tap referral/call actions.

Teleconsultation Live · 03:41

Bhavanipur Sub-Centre → Community Health Centre, Devgarh

Encrypted call

DR

Dr. Anjali Verma

MBBS · Medical Officer, CHC Devgarh

SD

Sunita Devi (ASHA) — camera off

PATIENT BEING DISCUSSED

RK

P008 · Female, 21 yrs
Case referred 10:42 AM · Bhavanipur

Emergency Escalation · Risk 10/11

PULSE
136 bpm

BP
173 mmHg

SPO2
93.4%

TEMP
100.0°F

CONSULTATION NOTES

10:44 — Shared vitals & symptom summary with Dr. Verma.

10:45 — Doctor advising ECG on arrival; suspects cardiac origin for chest pain.

10:46 — Ambulance (108) requested; ETA 18 minutes.

Add Doctor's Instructions

Track 108 Ambulance

Live demo 5 — Teleconsultation screen: ASHA worker on a live encrypted call with the PHC doctor for Case #P008, with vitals and consultation notes side by side.